Week 5 — Next Best Action Field Guide

Response vs. uplift · the three components of an NBA engine · affinity prediction by collaborative filtering · uplift estimators and the Qini curve · saturation, caps and substitution · the scoring layer · offline policy evaluation · bandits · patient predictive modeling and governance · orchestration · vocabulary · where the terminology comes from

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Reference handout · keep it for the term project and for Week 6 onward

Companion to Week_05_NBA_Notebook.ipynb and the Week 5 slides

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The concepts, formulas and decisions of a Next Best Action build, in the order a real build meets them — written for someone who has to defend the program to a brand team, a field director and a compliance officer in the same meeting.

Part 1 · Response vs. uplift — the distinction everything rests on

Response is what happens after an action: NBRx in the two weeks after a call. **Uplift** (the incremental effect, the conditional average treatment effect, CATE) is what happens after the action **minus what would have happened anyway**. A response model is a prediction problem; an uplift model is a causal problem. They have different targets, different data requirements and different evaluations — and the same vendors sell both under the same name. In pharma the model family is called **promotion response modeling** (Axtria, PMSA webinar); the four customer types below are Radcliffe & Surry's (Stochastic Solutions, 2011), the originating uplift-modelling source.

	Prescribes if NOT touched	Does not prescribe if NOT touched
Prescribes if touched	Sure things — uplift ≈ 0. A response model ranks them first. Spend here is wasted.	Persuadables — the only HCPs worth the cost of a touch.
Does not prescribe if touched	Sleeping dogs — uplift < 0. The touch pushes them away (over-touched competitor loyalists).	Lost causes — nothing moves them. Leave them alone.

The trap, in tonight's numbers

On the legacy arm (calls targeted by decile), called HCP-weeks show +1.06 NBRx vs. un-called. On the randomized arm the honest average effect of a call is +0.10 NBRx — $120 of margin for a $180 call.

A response model fit to the legacy arm puts 62% Core and 0% Entrenched in its top decile. It has learned the targeting rule and called it response.

Why it is a trap and not a bug: no amount of modeling on observational rows removes it. The fix is a different target (the difference) and a design that identifies it (randomization or a credible quasi-experiment).

Part 2 · The three components of an NBA engine

An NBA engine is three components in sequence. IQVIA's description of its own engine names them: “dynamic micro-segmentation and AI/ML-based scoring” on “a comprehensive 360-degree view of HCP preferences”.

Component	Question	Input	Method	Output	Tonight
1 · Segmentation / micro-segmentation	Who is this HCP?	Rx, patient mix, switching, channel response	k-means, two-stage; supervised growth score (Week 4)	Segment and growth group per HCP	Inherited from Week 4
2 · Affinity prediction	Which channels reach them?	Observed engagement on a subset of HCP × channel cells; a preference survey	Collaborative filtering (matrix factorization) + the segment prior for cold start	Affinity per HCP per channel — the full matrix	Notebook §4; Part 2b
3 · NBA layer	Which action pays this week?	Randomized touch log; business rules; affinities; value per script	Eligibility → uplift scoring → selection with exploration	One action per HCP per week, with its probability	Notebook §5–8; Parts 3–7

The NBA layer's three steps

Step	What it does	Who owns it
Eligibility (business rules)	For each HCP this week, the set of allowed actions. 'None' is always allowed. Consent, frequency caps, recency, sequencing, complaint and medical-information holds, the affinity floor from component 2.	Compliance, Privacy, the Field, Medical
Scoring	For each allowed action a: $\hat{\tau}_a(x)$ = expected incremental outcome given the context x; value = $\hat{\tau}_a(x)$ × value per unit − cost(a).	Analytics
Selection	argmax over allowed actions — with an exploration share (ε of HCP-weeks randomized) or a Thompson-sampling layer, and the chosen probability written to the log.	Analytics + Omnichannel

The exploration arm (a randomized test cell). In industry language this is a test-and-control design — ZS calls test and control “the backbone of success measurement”, McKinsey's omnichannel model refines content “through A/B testing in the field”, and IQVIA puts “local experimentation” in the execution layer. Tonight: a share of HCP-weeks (20–25%) in which the action is chosen at random among the eligible actions, regardless of who the HCP is — and the probability is logged. In bandit language it is the exploration arm. It is Week 3's geo-lift test at the HCP-week grain, and it is the data asset: the uplift model (Part 3), the fatigue curve (Part 4) and offline policy evaluation (Part 6) all come from it. The most common way NBA programs die is turning it off after the first good quarter.

Context (the x). Everything known at decision time: the Week 4 segment and growth group, decile, trailing NBRx, touches in the trailing 4 weeks by channel, weeks since last call, last action, consent, access (reject rate), specialty. Nothing from after the decision — the segmentation's feature-window discipline applies.

Part 2b · Affinity prediction — filling a matrix you only partly observe

The problem. The brand knows an HCP's affinity for a channel only where it has history on that HCP × channel, plus a few hundred surveyed HCPs. Tonight: 44% of the 6,000 × 6 cells, skewed to high-decile HCPs on call and email; 404 HCPs with no observed channel. The engine needs every cell — including the channels nobody has tried on that HCP, which is where it most wants to go.

The method. Collaborative filtering: HCPs who engage alike on the channels we have seen will engage alike on the ones we have not. Matrix factorization with biases (Koren, Bell & Volinsky, 2009 — the Netflix-prize method), fit on the residual from the segment × channel prior:

$$\text{affinity}(i, j) \approx \text{prior}(\text{segment of } i, j) + \text{bias}(i) + u(i) \cdot v(j)$$

u(i) is two numbers per HCP (roughly people ↔ digital, promotional ↔ scientific), v(j) two numbers per channel learned from everyone. Alternating ridge least squares; strong shrinkage on the HCP side (one or two known cells), almost none on the channel side (thousands). Cold start — an HCP with no known cell — reverts to the segment prior. The hybrid (matrix factorization where there is history, the prior where there is none) is what ships.

RMSE on affinity (0–1)	Held-out cells	HCPs with ≥ 3 known cells	Cold-start HCPs	Correlation with the full (planted) matrix
Segment × channel mean	0.184	0.182	0.185	0.55
kNN on Week 4 features	0.188	0.190	0.189	0.53
Matrix factorization	0.161	0.131	0.290	0.55
Hybrid (MF + prior)	0.146	0.131	0.185	0.68

- **What to take from the table:** the segment prior is a strong baseline — that is what a good micro-segmentation buys; collaborative filtering improves with the number of known channels; kNN on the Week 4 features adds nothing because the features are the segmentation.
- **Validation without truth:** hold out 20% of known cells and all cells of 10% of HCPs; and check predicted affinity against the engagement rate in the touch log (r = 0.20 / 0.29 / 0.33 on call / email / digital — modest, as a noisy binary over a handful of touches should be).
- **Two uses downstream:** the segment × channel view the brand team reads, and the affinity floor in the NBA layer — a channel with predicted affinity below 0.15 is not offered (email blocked for 34% of HCPs, digital 25%, call 15%).
- **Stated vs. revealed:** the survey measures what HCPs say (IQVIA Channel Dynamics); the history measures what they did. IQVIA's 2018 finding is that the two disagree more often than they agree. Tonight's matrix mixes both; a production engine keeps them as separate inputs.
- **Affinity ≠ uplift.** Core has the highest affinity on five of six channels and one of the lowest returns per dollar. Affinity answers how to reach; uplift answers whether it pays.

Part 3 · Uplift estimators, and how to evaluate one

Target: $\tau_a(x) = E[Y \mid X = x, A = a] - E[Y \mid X = x, A = \text{none}]$, estimated on the randomized rows only, with the hold-out split by HCP so that it contains prescribers the model never saw.

Estimator	How	Strength	Weakness	Tonight (Qini: call / email / digital)
S-learner	One outcome model with the action as a feature; read the effect off by flipping the feature	Simple; uses every row	Shrinks the effect toward zero — the action is one feature among forty	133 / 75 / 74
T-learner	One outcome model per arm; subtract	Honest; no shrinkage	Each arm's model sees only its rows; small effects drown in noise	174 / 51 / 18
X-learner	T-learner; then model the imputed individual effects on each arm and blend with propensity weights	Built for unequal arms (90% control is typical); models the effect directly	More moving parts	192 / 76 / 62 — the production model
Causal forest	Honest trees that split on the effect (Athey & Imbens; econml CausalForestDML)	Valid confidence intervals on τ(x)	Slow; a dependency; same data limits	optional (BAYES-style flag)

Evaluation without individual labels. No HCP-week has an observed uplift, so uplift models are evaluated on groups: sort the hold-out by predicted uplift, cut into deciles, and inside each decile compare the randomized treated and control means. The **Qini curve** cumulates (treated mean − control mean) × treated count along the ranking; the area above the random diagonal (the Qini coefficient) is the one-number summary. A model that is good at finding persuadables rises steeply and then flattens. R^2, AUC and log-loss say nothing about uplift and must not be quoted.

What tonight's grade against the planted truth says (instructor file)

Correlation of predicted with planted uplift on hold-out HCP-weeks: call 0.59, email 0.39, digital 0.56. The ordering that matters is right: Movers first for the call and the digital placement, Entrenched negative everywhere, Long tail ≈ 0.

Core is over-estimated (sure things with a big baseline look incremental to a count model). The single best channel per HCP-week is named 41% of the time (chance 33%).

26 weeks of a 25% exploration arm is about six randomized weeks per HCP. That is the real-world budget too. An NBA engine is a segment-and-context machine with individual refinements — which is why the rules and the segment priors matter as much as the model.

Part 4 · Saturation and fatigue, frequency caps, channel substitution, consent

IQVIA has documented the phenomenon for years: in the most digitally promoted markets “the majority of HCP groups would prefer less communication via digital channels”, and the initial success of digital “resulted in overkill and saturation” (Channel Preference Versus Promotional Reality, 2018); its Channel Dynamics work tracks “digital fatigue” (2023). The uplift in Part 3 is an average over states. The marginal touch has its own curve, estimated on the randomized rows with a Poisson regression and each HCP's own baseline (log of 2 × trailing weekly NBRx) as the offset. Tonight's estimates:

Touches already delivered in the trailing 4 weeks	0	1	2	3	4
Lift of one more touch — all segments except Entrenched	+27%	+18%	+11%	+8%	−5%
Lift of one more touch — Entrenched (sleeping dogs)	−3%	−30%	−22%	−15%	−22%

- **Frequency cap:** 3 touches per 4 weeks; 2 for Entrenched. Estimated, not guessed — and re-estimated every refresh. ('Frequency cap' is the digital-advertising term; in pharma orchestration it is one of the business rules that, in Axtria's phrase, combine with the AI/ML models and governance 'to drive precision targeting'.)
- **Channel substitution:** an email after a quiet week +10%; the week after a rep call +2% (interaction p = 0.015). Rule: no email within a week of a call.
- **Consent:** 2,450 emails to HCPs without consent produced 135 complaints and no lift. Rule: hard gate, owned by Privacy.
- **Belt and braces:** the uplift model also sees touches_4w and last_action as features, so fatigue and substitution are inside $\hat{\tau}$ — and the rules still run in front of it, because the field needs a rule it can quote and compliance needs one it can audit.

Part 5 · The scoring layer — value, cost, and 'none'

value(a | x) = $\hat{\tau}_a(x) \times V - \text{cost}(a)$, with value(none) = 0, and the recommendation is the argmax over the eligible set. Tonight's conventions, which you should be able to defend or replace: V = $1,200 per incremental NBRx (≈ 4.8 fills × $250 gross margin per TRx, from the Week 1 P&L via Week 3's MMM); cost = $180 per rep call, $2 per approved email, $6 per digital placement.

Consequence	Number	Why it matters
Break-even uplift for a call	0.15 NBRx per 2 weeks	The average call (+0.10) loses money; only persuadables clear it
Calls per week, legacy → NBA	955 → 297	The field will read this as a headcount question — bring the OPE interval
HCPs recommended 'none' next week	60%	The honest output of an engine whose cost line is real
Model-estimated net value next week	$838k (NBA) vs $169k (legacy mix)	'Model-estimated' — the number that justifies Part 6, not a rollout

Part 6 · Offline policy evaluation (OPE)

How to score a policy that was never run, on a log that was. All three estimators need the logging propensity $p(a_t \mid x_t)$ for every row — the probability with which the action actually taken was chosen. Write the rules engine as probabilities; log them.

Estimator	Formula	Property
IPS	$\hat{V}(\pi) = (1/n)\ \Sigma_t\ [\pi(a_t \mid x_t) / p(a_t \mid x_t)] \cdot r_t$	Unbiased; high variance when π takes actions the logger rarely took; clip the weights
SNIPS	$\Sigma_t\ w_t\ r_t\ /\ \Sigma_t\ w_t$	Slightly biased; much lower variance; invariant to reward shifts
Doubly robust	$(1/n)\ \Sigma_t\ [\ \Sigma_a\ \pi(a \mid x_t)\ \hat{r}(x_t,a) + w_t\ (r_t - \hat{r}(x_t,a_t))\ ]$	Unbiased if either the propensities or the reward model is right

Tonight's OPE (net value per HCP-week; observed $1,378)

Logging policy $1,378 (sanity check passes) · uniform random $1,347–1,412 · call deciles 9–10 $1,397–1,407 · response-model policy $1,403–1,415 · uplift NBA + suppression $1,463–1,483 (IPS ± 2 s.e. ≈ ± $32). All three estimators agree on the ordering.

Effective sample size is the price: the NBA policy is scored on ≈ 37,000 effective rows of 156,000. A policy far from the logger is evaluated on the few rows that overlap.

What it is for: justifying the A/B test. What it is not for: rolling out. The brand decides on the A/B (a 90/10 split with a global holdout), not on the OPE.

Part 7 · Contextual bandits — learning the policy online

An uplift model + argmax is a one-shot policy. A contextual bandit keeps learning: each week it picks an action given the context, observes the reward, updates. The tension is explore/exploit.

- **ε-greedy:** act on the best current estimate, explore a fixed share ε. Learns in weeks; pays the exploration tax forever (regret grows linearly, slope ∝ ε).

- **Thompson sampling:** sample from the posterior of each action's value and act greedily on the sample. Explores where uncertain, stops where not; regret flattens. Tonight: regret $282 per HCP over 52 weeks vs $1,403 (ε = 0.10) and $11,848 (fixed mix).
- **LinUCB / linear Thompson:** the production form — a linear reward model on the context, with an optimism or posterior term per action. The 'arms' are the actions scored by the uplift model; the fatigue state is in the context; the cap removes over-touch actions from the arm set.
- **Offline first:** a bandit is evaluated offline with the same estimators as Part 6 (replay), before it is allowed to choose actions for real HCPs.

Part 8 · Patient predictive modeling — the patient-side NBA, and its governance

The same engine pointed at patients: which patients on therapy will discontinue, early enough to do something — PSP enrollment outreach via the prescriber's office, a copay-support reminder, an adherence nudge. Three disciplines:

1. **Only allowed features.** Journey and access features (PDC, refill gaps, copay, PSP, portal, prior lines, comorbidity count, ER visits, payer type, prescriber specialty). No free text, no ZIP, no dates of birth, no sensitive attributes. Purpose limitation decides the list, not the leaderboard.
2. **Evaluate for the decision.** Tonight: AUC 0.77 — and the number that matters is that the top two risk deciles discontinue at 40% (base 18%) and hold 45% of all discontinuers. That is a nurse's call list.
3. **Fairness before launch.** Calibration and precision by payer type and by prescriber specialty, with a documented threshold for 'not equal enough to ship'. Tonight Commercial, Medicare Part D and Medicaid sit on the diagonal together.

Control	What it means in practice
Purpose limitation	The score supports adherence; it is never used to deprioritize a patient, a prescriber or a payer
Data minimization	The feature list above and nothing else; outputs aggregated or pseudonymized before they leave the modeling environment
Consent for the channel	PSP outreach needs PSP consent; the prescriber's office is the compliant default
Fairness check	Calibration by payer and specialty, re-checked at every refresh
Drift monitoring	Score distribution, outreach mix, calibration — monthly, with an owner
Human in the loop	The nurse decides, the model ranks; the override rate is a model metric

Part 9 · Orchestration — triggers, suppression rules, the CDP, measurement

- **Signals and triggers:** events that start a decision — an NBRx jump in a Movers HCP (reinforce), a reject rate crossing 25% (switch to access content), a Week 4 exit trigger (re-segment), a patient's refill gap past 30 days (patient NBA). Weekly batch for the field; event-driven for digital.
- **Business rules (suppression rules):** Part 4's rules plus the compliance holds; they run before scoring; they are owned outside analytics. Axtria's orchestration framework names the three ingredients in that order — AI/ML models, business rules, governance.
- **Suggestions and acceptance:** the field-facing form of an NBA is a suggestion in CRM (Veeva CRM Suggestions; Axtria Triggers). IQVIA reports tracking “acceptance of suggestions and follow through”; suggestion acceptance is a program KPI — it says whether the field trusts the engine, not whether the engine works. Axtria and ZS both now argue for explainable suggestions and a 'closed loop' in which every accepted or ignored suggestion is a learning event.

- **The CDP:** the customer data platform owns identity (one HCP across CRM, email, web, Rx) and the consent record. IQVIA's phrase is the 'common data spine' (reading pack, Part C). Without it the NBA engine is a spreadsheet.
- **Measurement:** a standing 10% global holdout; the exploration arm inside the treated population; OPE every quarter; a quarterly A/B of engine versions. Report incremental NBRx and net value — never engagement.

Part 10 · Vocabulary

Term	Meaning
Next Best Action (NBA)	A per-customer, per-period recommendation of one action — channel and content, including 'none' — chosen to maximize expected incremental value under business rules (IQVIA, Veeva, EY, McKinsey)
Suggestion	The field-facing form of an NBA: a recommendation delivered to the rep in CRM, with a reason (Veeva CRM Suggestions; Axtria Triggers)
Next Best Experience (NBX)	The 2025–26 evolution of NBA: orchestrated, contextual experiences across every touchpoint and stakeholder, with closed-loop learning (ZS; Axtria)
Promotion response (model)	The pharma name for models of the impact of promotional activity on prescribing — the family tonight's uplift model belongs to (Axtria)
Uplift / CATE / heterogeneous treatment effect	The incremental effect of an action for a customer with context x; the target of an uplift model
Response model / propensity-to-prescribe	A prediction of the outcome after an action, from observational rows; not causal
Exploration arm	The share of decisions chosen at random among eligible actions, with the probability logged
Logging propensity	p(a | x): the probability with which the action actually taken was chosen; required for OPE
S-/T-/X-learner, causal forest	Meta-learners and a tree method for estimating uplift from randomized (or well-adjusted) data
Qini curve / coefficient	The uplift evaluation: cumulative incremental outcome along the predicted-uplift ranking; area above random
Saturation / fatigue / frequency cap	The declining (then negative) effect of the nth touch in a window (IQVIA: 'overkill and saturation', 'digital fatigue'), and the business rule that stops before it
Test and control / holdout	The measurement design pharma trusts: a treated cell and a comparable untreated cell, randomized where possible (ZS; McKinsey)
Sleeping dogs	Customers for whom the touch has a negative effect
Channel substitution	A touch's effect reduced by a recent touch on another channel
Channel preference / affinity	Stated (surveys: IQVIA Channel Dynamics) or revealed (engagement) per channel — decides how to reach a customer, not whether it is worth it
Affinity prediction / collaborative filtering	Filling the unobserved HCP × channel cells from the observed ones by matrix factorization (Koren et al.), with the segment prior for cold start
Micro-segmentation	The Week 4 segments refined with specialty and access — the prior for every HCP the engine has not yet learned about (IQVIA: 'dynamic micro-segmentation')
Cold start	An HCP (or channel) with no observed history — predicted from features and segment, not from behaviour
Engagement vs. impact	Activity and adoption metrics (opens, clicks, calls, suggestion acceptance) are a proxy; the outcome is incremental prescribing and net value (ZS)
Closed-loop learning	Every interaction and every accepted or ignored suggestion is a learning event for the next model (ZS; Axtria)
IPS / SNIPS / doubly robust	Offline policy evaluation estimators that re-weight logged rewards by $\pi(a\|x)/p(a\|x)$
Effective sample size	$(\Sigma w)^2 / \Sigma w^2$ — how many logged rows a target policy is really evaluated on
Contextual bandit	An online policy that chooses actions given context and learns from the rewards; ε-greedy, Thompson sampling, LinUCB
Regret	The cumulative gap between the reward earned and what the best action in each context would have earned
Global holdout	A share of customers kept outside the engine to measure its total incremental effect

Single customer view / customer 360 / CDP	One identity, consent record and touch history per HCP across channels (IQVIA; McKinsey); the customer data platform is the system that holds it; IQVIA's 'common data spine'
Human in the loop / guardrails	The rep or nurse decides, the engine ranks; the enterprise sets the boundaries (McKinsey; ZS; IQVIA)
PSP	Patient support program — the brand's adherence, copay and nurse services; the patient NBA's main channel

Part 11 · Where the terminology comes from

Term → sources, then the citations with what tonight takes from each. 'In the course library' = the IQVIA PDFs in the reading folder; the rest are in the readings file. Contextual bandit and offline policy evaluation come from the online-advertising literature, not a pharma publication.

Term	How tonight uses it	Sources
Next Best Action (NBA)	A per-HCP, per-period recommendation of one action — channel and content — including 'none'	IQVIA 2025; Veeva Systems 2015; EY 2023; McKinsey 2022
Suggestions / triggers / signals	The field-facing form of NBA (a suggestion in CRM) and the events that start one (a diagnosis, a formulary change, a reject-rate crossing)	Veeva Systems 2015; Axtria 2020; ZS 2024; IQVIA 2026
Next Best Experience (NBX)	The 2025–26 evolution: from isolated suggestions to orchestrated, contextual experiences across stakeholders	ZS 2026; Axtria 2025
Omnichannel orchestration	Coordinating personal and non-personal channels into one sequence per HCP, with outcomes logged	IQVIA 2026; IQVIA 2024; Axtria 2025; McKinsey 2022
Micro-segmentation	The Week 4 segments, refined with specialty and access — the prior for every HCP the engine has not yet learned about	IQVIA 2025; McKinsey 2022
Channel preference / affinity; affinity prediction (collaborative filtering)	Which channels an HCP engages with — stated (surveys) and revealed (engagement); observed for a subset, predicted for the rest by matrix factorization	IQVIA 2024; IQVIA 2018; IQVIA 2023; IQVIA 2025; Koren, Bell & Volinsky, IEEE Computer 2009
Saturation / fatigue; frequency cap	The declining then negative effect of the nth touch; the business rule that stops before it ('frequency cap' is the digital-advertising term adopted by pharma orchestration)	IQVIA 2018; IQVIA 2023; Axtria 2025
Business rules / suppression rules	Consent, caps, recency, holds — run before scoring, owned outside analytics	Axtria 2025; IQVIA 2026
Promotion response	The impact of promotional activity on prescribing — the model family tonight's uplift model belongs to	Axtria 2021; ZS 2024
Uplift / incremental lift; persuadables, sure things, lost causes, sleeping dogs	The effect of an action minus what would have happened anyway, and the four customer types it separates	Stochastic Solutions (N. Radcliffe & P. Surry) 2011; Athey & Imbens, PNAS 2016
Test and control; holdout; randomized test cell (the 'exploration arm')	The measurement design: a randomized share of decisions, and a share of HCPs outside the engine	ZS 2026; McKinsey 2022; IQVIA 2026
Engagement vs. impact	Activity and adoption metrics are a proxy; the outcome is incremental prescribing and net value	ZS 2026; IQVIA 2025
Single customer view / customer 360 / common data spine / CDP	The identity-and-consent layer under the engine	IQVIA 2026; IQVIA 2024; McKinsey 2022; EY 2023
Closed-loop learning; suggestion acceptance	Every interaction and every accepted/ignored suggestion feeds the next model; acceptance rate is a program KPI, not an outcome	ZS 2026; IQVIA 2025; Axtria 2025
Human in the loop; guardrails	The rep and the nurse decide; the engine ranks; the enterprise sets the boundaries	McKinsey 2025; ZS 2026; IQVIA 2025
Patient journey; closed patients; provider behavior	Longitudinal patient-level data read at the provider level — the Week 2/4 foundation of the patient NBA	Komodo Health 2023; IQVIA 2025
Contextual bandit; offline policy evaluation (IPS, SNIPS, doubly robust)	Online learning of the policy, and scoring a policy from a log. Academic / online-advertising vocabulary, labelled as such	Li, Chu, Langford & Schapire (WWW); Dudík, Langford & Li (ICML) 2010

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